Lab Programs

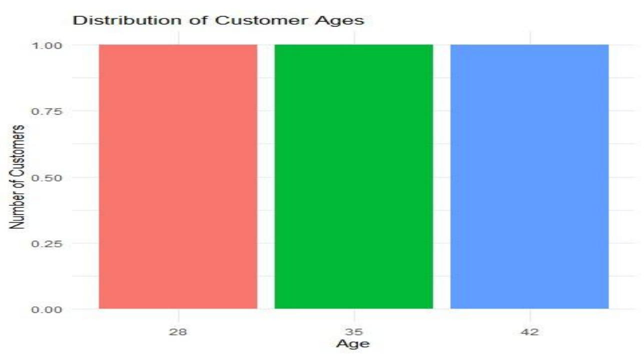
SET 7 – Customer Demographics Analysis

7A. Bar Chart

Program

```
library(ggplot2)

customer<-data.frame(
Age=c(28,35,42)
)

ggplot(customer,aes(factor(Age)))+
geom_bar(fill="steelblue")+
labs(title="Customer Age Distribution",
x="Age",
y="Count")+
theme_minimal()
```

Output

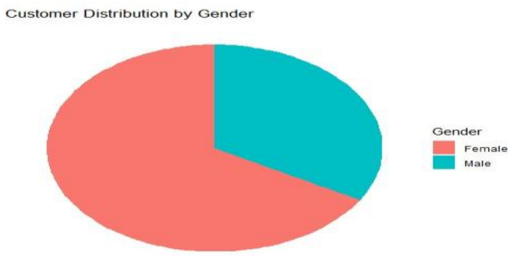


7B. Pie Chart

Program

```
library(ggplot2)

gender<-data.frame(
Gender=c("Female","Male","Female")
)

ggplot(gender,aes(x="",fill=Gender))+
geom_bar(width=1)+
coord_polar("y")+
labs(title="Gender Distribution")+
theme_void()
```

Output



7C. Table

Program

```
customer<-data.frame(
ID=1:3,
Age=c(28,35,42),
Gender=c("Female","Male","Female"),
Income=c(50000,60000,75000)
)

print(customer)
```

Output

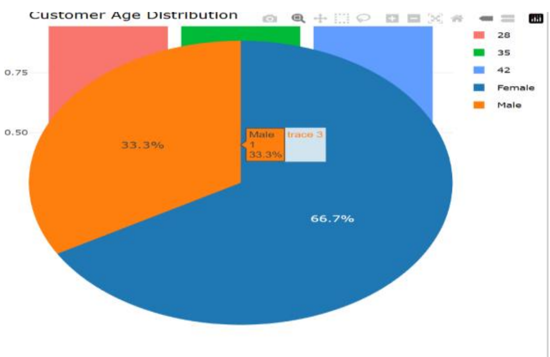
```
Table: Customer Demographics Information

| CustomerID| Age|Gender | Income|
|----------:|---:|:------|------:|
|          1|  28|Female |  50000|
|          2|  35|Male   |  60000|
|          3|  42|Female |  75000|
>
```

7D. Interactive Dashboard

Program

```
library(ggplot2)
library(plotly)

customer<-data.frame(
Age=c(28,35,42),
Gender=c("Female","Male","Female")
)

p1<-ggplot(customer,aes(factor(Age)))+
geom_bar()

p2<-ggplot(customer,aes(x="",fill=Gender))+
geom_bar(width=1)+
coord_polar("y")

subplot(ggplotly(p1),ggplotly(p2),nrows=2)
```

Output

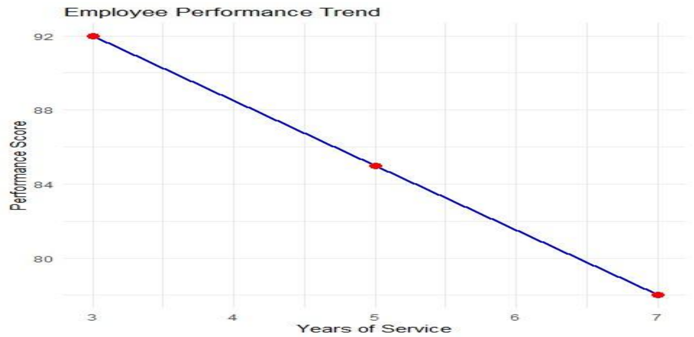


SET 8 – Employee Performance Analysis

8A. Line Chart

```
library(ggplot2)

emp<-data.frame(
Years=c(5,3,7),
Score=c(85,92,78)
)

ggplot(emp,aes(Years,Score))+
geom_line(color="blue")+
geom_point(size=3,color="red")+
labs(title="Employee Performance",
x="Years",
y="Score")+
theme_minimal()
```

Output



8B. Bar Chart

```
library(ggplot2)

emp<-data.frame(
Department=c("Sales","HR","Marketing")
)
```

ggplot(emp,aes(Department))+

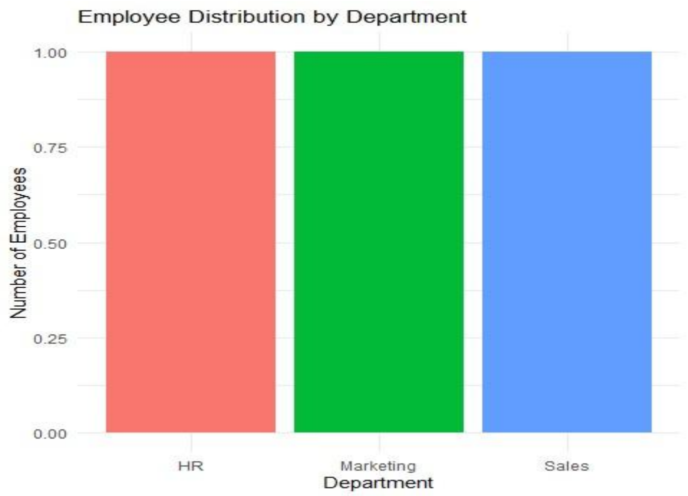
geom_bar(fill="orange")+

labs(title="Department Distribution",

x="Department",

y="Employees")

Output



8C. Table

emp<-data.frame(

ID=1:3,

Department=c("Sales","HR","Marketing"),

Years=c(5,3,7),

Score=c(85,92,78)

)

print(emp)

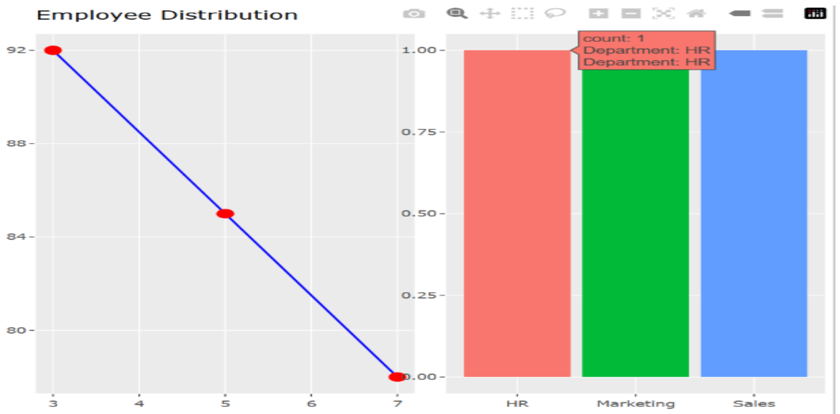
Output

```
Table: Employee Performance Data

| EmployeeID|Department | YearsOfService| PerformanceScore|
|----------:|:----------|--------------:|----------------:|
|          1|Sales      |              5|               85|
|          2|HR         |              3|               92|
|          3|Marketing  |              7|               78|
>
```

8D. Interactive Dashboard

```
library(ggplot2)
library(plotly)

emp<-data.frame(
Department=c("Sales","HR","Marketing"),
Years=c(5,3,7),
Score=c(85,92,78)
)

p1<-ggplot(emp,aes(Years,Score))+geom_line()+geom_point()

p2<-ggplot(emp,aes(Department))+geom_bar()

subplot(ggplotly(p1),ggplotly(p2),nrows=2)
```

Output



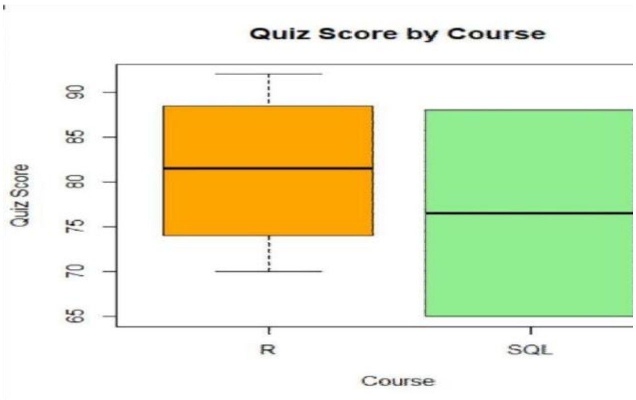
SET 9 – Online Learning Activity Analysis

9A. Histogram & Boxplot

Program

```
library(ggplot2)

data <- data.frame(
Course=c("R","R","SQL","R","R","SQL"),
```

```
Quiz_Score=c(78,85,65,92,70,88)
)

ggplot(data,aes(Quiz_Score))+
geom_histogram(binwidth=5,fill="skyblue",color="black")+
labs(title="Quiz Score Distribution")

ggplot(data,aes(Course,Quiz_Score,fill=Course))+
geom_boxplot()+
labs(title="Quiz Score by Course")
```



Output

9B. Scatter Plot

Program

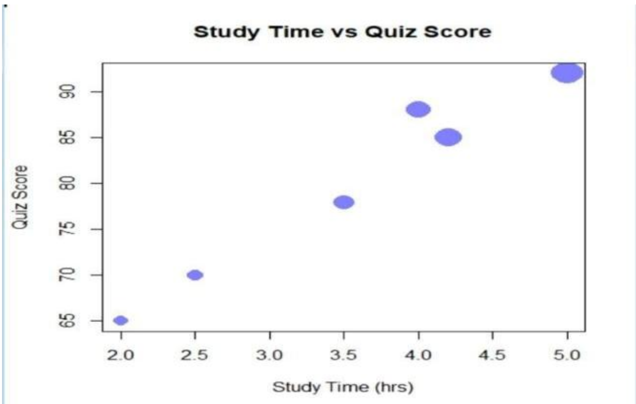
```
library(ggplot2)

data<-data.frame(
Study_Time=c(3.5,4.2,2.0,5.0,2.5,4.0),
Quiz_Score=c(78,85,65,92,70,88),
Videos=c(12,15,8,18,9,14)
)

ggplot(data,
```

aes(Study_Time,Quiz_Score,size=Videos))+

geom_point(alpha=0.7,color="blue")+

labs(title="Study Time vs Quiz Score")

Output

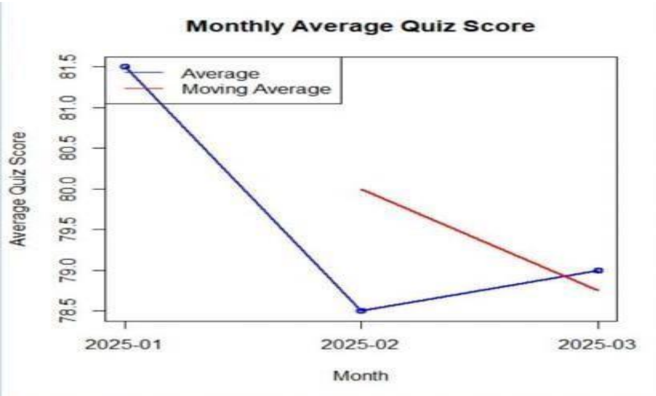


9C. Line Chart

Program

```
library(ggplot2)

data<-data.frame(
Month=c("Jan","Feb","Mar"),
Average=c(81.5,78.5,79)
)

ggplot(data,aes(Month,Average,group=1))+
geom_line(color="red")+
geom_point(size=3)+
labs(title="Average Monthly Quiz Score")
```

Output

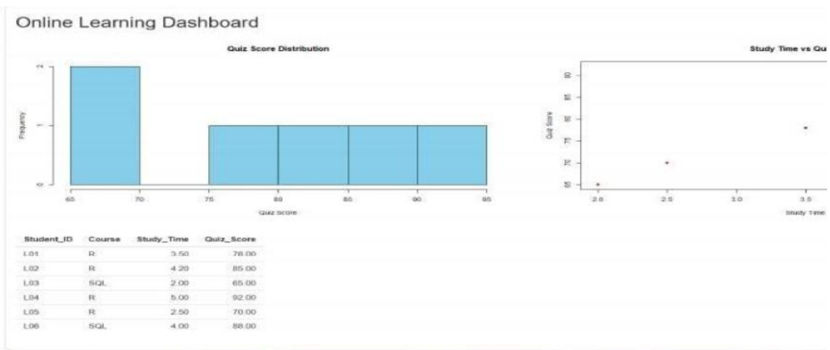


9D. Tableau Dashboard (R Interactive)

Program

```
library(ggplot2)
library(plotly)

data<-data.frame(
Course=c("R","SQL"),
Average=c(81.25,76.5)
)

p1<-ggplot(data,aes(Course,Average,fill=Course))+
geom_col()

p2<-ggplot(data,aes(Course,Average))+
geom_point(size=4)

subplot(ggplotly(p1),ggplotly(p2),nrows=2)
```

Output



Student_ID	Course	Study_Time	Quiz_Score
L01	R	3.50	78.00
L02	R	4.20	85.00
L03	SQL	2.00	65.00
L04	R	5.00	92.00
L05	R	2.50	70.00
L06	SQL	4.00	88.00

SET 10 – Survey Responses Analysis

10A. Grouped Bar Chart

```
library(ggplot2)

survey<-data.frame(
Answer=c("A","B","C"),
Count=c(1,1,1)
)

ggplot(survey,aes(Answer,Count,fill=Answer))+
geom_col()+
labs(title="Question 1 Responses")
```

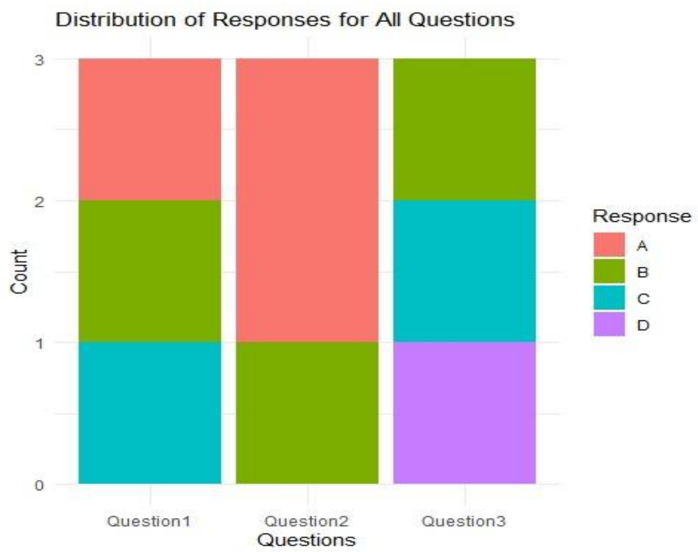
Output

10B. Stacked Bar Chart

```
library(ggplot2)
library(reshape2)

data<-data.frame(
Question=c("Q1","Q2","Q3"),
A=c(1,2,1),
B=c(1,1,1),
```

```
C=c(1,0,1)
)

long<-melt(data,id="Question")

ggplot(long,aes(Question,value,fill=variable))+
geom_col()+
labs(title="Survey Responses")
```

Output



10C. Table

```
survey<-data.frame(
ID=1:3,
Q1=c("A","B","C"),
Q2=c("B","A","A"),
Q3=c("C","D","B")
)

print(survey)
```

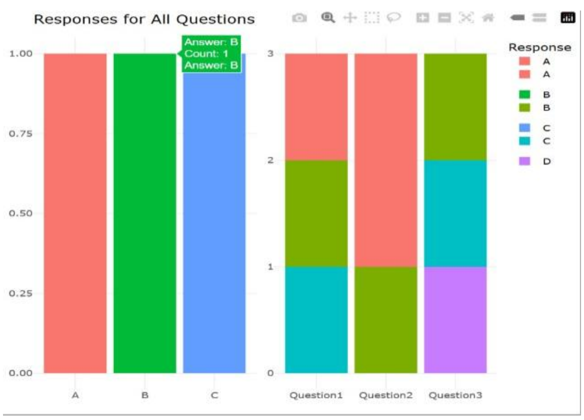
Output

```
Table: Survey Response Data

| SurveyID|Question1 |Question2 |Question3 |
|--------:|:---------|:---------|:---------|
|        1|A         |B         |C         |
|        2|B         |A         |D         |
|        3|C         |A         |B         |
>
```

10D. Interactive Dashboard

```
library(ggplot2)

library(plotly)


survey<-data.frame(

Answer=c("A","B","C"),

Count=c(1,1,1)

)


p1<-ggplot(survey,aes(Answer,Count))+geom_col()


p2<-ggplot(survey,aes(Answer,Count))+geom_point(size=4)


subplot(ggplotly(p1),ggplotly(p2),nrows=2)
```

Output

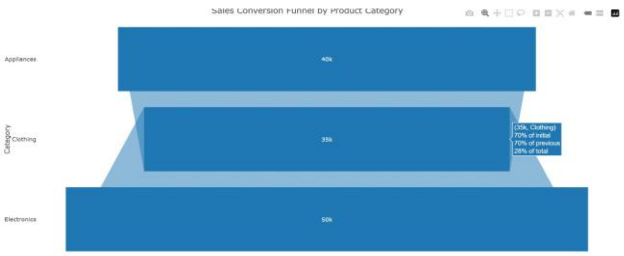


SET 11 – Product Category Analysis

11A. Funnel Chart

```
library(ggplot2)

category<-data.frame(
Category=c("Electronics","Clothing","Appliances"),
Sales=c(50000,35000,40000)
)

ggplot(category,aes(reorder(Category,Sales),Sales,fill=Category))+
geom_col()+
coord_flip()+
labs(title="Sales Funnel")
```

Output



11B. Table

```
category<-data.frame(
Category=c("Electronics","Clothing","Appliances"),
Sales=c(50000,35000,40000)
)

print(category)
```

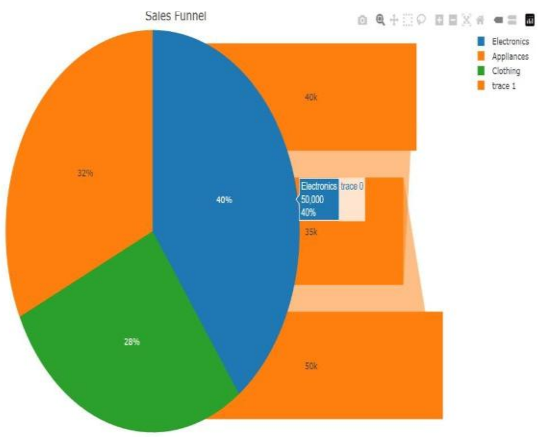
Output

```
Table: Product Category Sales

|Category    | Sales|
|:-----------|-----:|
|Electronics | 50000|
|Clothing    | 35000|
|Appliances  | 40000|
> |
```

11C. Interactive Dashboard

```
library(ggplot2)
library(plotly)

category<-data.frame(
Category=c("Electronics","Clothing","Appliances"),
Sales=c(50000,35000,40000)
)

p1<-ggplot(category,aes(Category,Sales,fill=Category))+geom_col()

p2<-ggplot(category,aes(x="",y=Sales,fill=Category))+
geom_col(width=1)+
coord_polar("y")

subplot(ggplotly(p1),ggplotly(p2),nrows=2)
```

Output

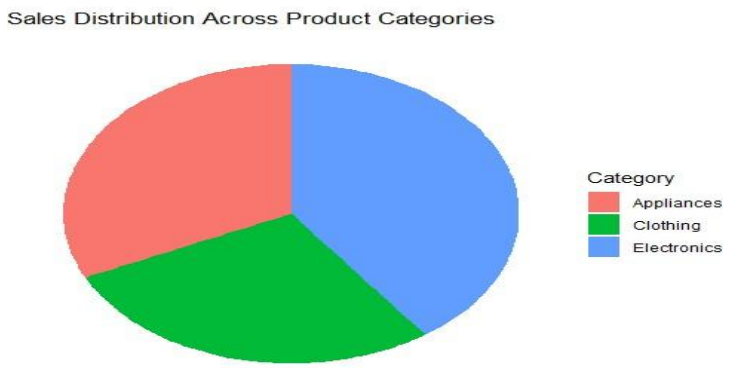


11D. Pie Chart

```
library(ggplot2)

category<-data.frame(
Category=c("Electronics","Clothing","Appliances"),
```

```
Sales=c(50000,35000,40000)
)

ggplot(category,aes(x="",y=Sales,fill=Category))+
geom_col(width=1)+
coord_polar("y")+
labs(title="Sales Distribution")
```

Output



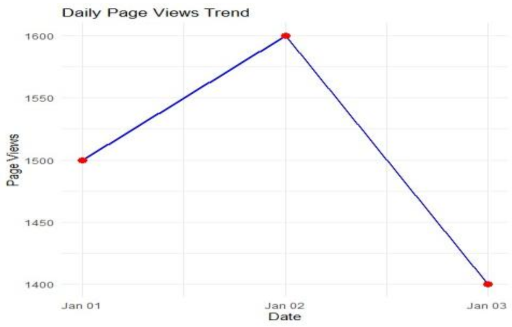
SET 12 – Website Traffic Analysis

12A. Line Chart

```
library(ggplot2)

web<-data.frame(
Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03")),
Views=c(1500,1600,1400)
)

ggplot(web,aes(Date,Views))+
geom_line(color="blue")+
geom_point(color="red")+
```

```
labs(title="Website Traffic")
```

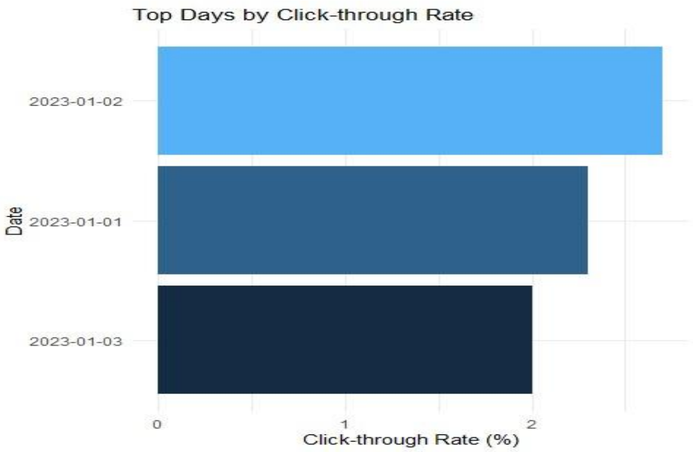
Output



12B. Bar Chart

```
library(ggplot2)

web<-data.frame(
Date=c("Jan1","Jan2","Jan3"),
CTR=c(2.3,2.7,2.0)
)

ggplot(web,aes(Date,CTR,fill=Date))+
geom_col()+
labs(title="Top CTR Days")
```

Output



12C. Table

```
web<-data.frame(

Date=c("2023-01-01","2023-01-02","2023-01-03"),

PageViews=c(1500,1600,1400),

CTR=c(2.3,2.7,2.0)

)


print(web)
```

Output

```
Table: Website Traffic Data

|Date        | PageViews| CTR|
|:-----------|---------:|---:|
|2023-01-01  |      1500| 2.3|
|2023-01-02  |      1600| 2.7|
|2023-01-03  |      1400| 2.0|
>
```

12D. Interactive Dashboard

```
library(ggplot2)

library(plotly)


web<-data.frame(

Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03")),

Views=c(1500,1600,1400),

CTR=c(2.3,2.7,2.0)

)


p1<-ggplot(web,aes(Date,Views))+geom_line()


p2<-ggplot(web,aes(Date,CTR))+geom_col()


subplot(ggplotly(p1),ggplotly(p2),nrows=2)
```

Output

Top CTR Days
Date: 2023-01-02
PageViews: 1600
group: 1
1600
1550
1500
1450
1400
Jan 01
Jan 02
Jan 03
2
1
0
2023-01-01
2023-01-02
2023-01-03